

TrES-2b

R-1479

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_140828 · created 2026-08-01T12:40:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456897.66488 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.013
Transit depth (Rp/Rs)^2	1.76 +/- 0.35 [%]
Orbital Inclination (inc)	83.66 +/- 0.94
Airmass coefficient 1 (a1)	1.0914 +/- 0.0065
Airmass coefficient 2 (a2)	-0.078 +/- 0.027
Scatter in the residuals of the lightcurve fit is	0.6 %
Best Comparison Star	None
Optimal Aperture	4.25
Optimal Annulus	9.55
Transit Duration (day)	0.067 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.013 vs 0.1254 (deviation 0.32σ)
Duration fit vs published	0.067 d vs 0.0746 d (deviation 0.23σ)
Mid-time O–C	-0.8 min
Depth SNR	5.7
Residual scatter	0.6 %

Light curve

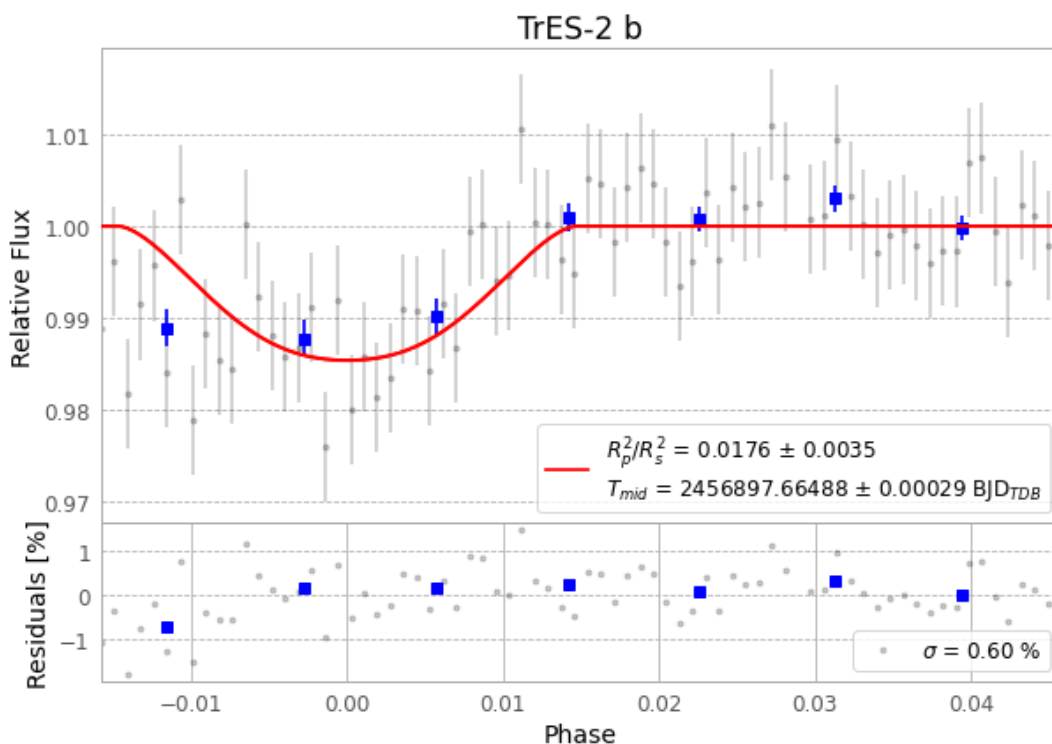


Figure 1 — FinalLightCurve_TrES-2 b_2014-08-27.png (SHA-256 82f22002a41c55fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [235, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7195b12d5684b00...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), TRES-2140828025712.FITS → TRES-2140828063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140828.log (fd435e625783...), FinalLightCurve_TrES-2 b_2014-08-27.png (82f22002a41c...), FinalLightCurve_TrES-2 b_2014-08-27.csv (8ecfb565b302...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:40Z.